Measures of Central Tendency and Variation

Questions 1–52

Question 1. For the two data sets below, find the mean, median, mode, range and interquartile range, and hence compare the two sets of data.

Set A	3	3	6	7	8	12	15	16	20	25	28
Set B	3	3	6	9	10	12	13	16	20	23	28

Question 2. Give one reason why:

a) the mean would be preferable to use as an average instead of the median,

b) the median would be preferable to use as an average instead of the mean.

Question 3. The following data was recorded:

1	4	8	8	9	10	11	14	15	15	25

a) Find the mean and standard deviation.
b) Values that are less than two standard deviations below or above the mean are considered outliers. Determine whether there are any outliers in the data.
c) Find the median and interquartile range.
d) Values that are less than LQ $-1.5\times$ IQR or more than UQ $+1.5\times$ IQR are considered outliers. Determine whether there are any outliers in the data.

Question 4. The following data was recorded:

3	8	9	10	15	17	21	25	26	28	29	55

a) Find the mean and standard deviation.
b) Values that are less than two standard deviations below or above the mean are considered outliers. Determine whether there are any outliers in the data.
c) Find the median and interquartile range.
d) Values that are less than LQ $-1.5\times$ IQR or more than UQ $+1.5\times$ IQR are considered outliers. Determine whether there are any outliers in the data.

Question 5. The following data was recorded:

35	36	45	48	52	59

a) Find the mean.
b) Without performing any calculations, explain what would happen to the mean if 59 was removed.

Question 6. The following data was recorded:

21	25	40	61	90	102

a) Find the mean.
b) Without performing any calculations, explain what would happen to the mean if 40 was changed to 56.5

Question 7. Find the mean, $\bar{x}$,

a) given that $\sum x = 860,\ n = 5$

b) given that $\sum x = 86753,\ n = 35$

Question 8. A data set, x, contains n values. Given that $\sum x = 51324$ and $\bar{x} = 52$, determine the value of n

Question 9. Given that $\sum x = 613, \sum x^2 = 56905, n = 10$, find the standard deviation, σ, to 3 significant figures.

Question 10. Given that $\sum x = 412, \sum x^2 = 7322, n = 25$, find the standard deviation, σ, to 3 significant figures.

Question 11. Given that $\sum x = 3305, \sum x^2 = 183075, n = 60$, find the variance, σ^2, to 3 significant figures.

Question 12. Given that $\sum x = 1354, \sum x^2 = 212182, n = 9$, find the variance, σ^2, to 3sf.

Question 13. The following data was recorded:

109	112	113	117	118	125

a) Find the standard deviation.
b) Without performing any calculations, explain what would happen to the standard deviation if 125 was removed.

Question 14. The following data was recorded:

37.4	39.2	40.1	40.3	42.1	43.6

a) Find the standard deviation.
b) Without performing any calculations, explain what would happen to the standard deviation if 43.6 was changed to 48.6

Question 15. The following data was recorded:

8	11	13	16	18	21

a) Find the mean.
b) Find the standard deviation.
c) Two more data points are added, x and y, such that $x < y$. It is given that the mean remains the same but the standard deviation is reduced. Determine possible values for x and y

Question 16. The following data was recorded:

21	25	32	34	38	42

a) Find the mean.
b) Find the median.
c) Two more data points are added, x and y, such that $x < y$. It is given that the mean remains the same but the median is reduced. Determine possible values for x and y

Question 17. The following data was recorded:

8	12	15	x	22	28

a) Given that the mode is 22, find the value of x
b) Given instead that the mean is 17.05, find the value of x
c) Given instead that the median is 16.3 and $15 < x < 22$, find the value of x

Question 18. Chrissy grows a certain variety of sunflowers. One day, she measures the heights, in cm, of a random sample of 12 of the sunflowers in her garden and her collected data is shown below.

85	102	121	127	132	133

140	145	149	160	162	198

a) Find the mean height of the sunflowers to 3 significant figures.
b) Find the sample standard deviation of the heights of the sunflowers to 3 significant figures.
c) Any sunflower with a height of less or more than 2 standard deviations from the mean may be regarded as an outlier. Using your answers to a) and b), determine if there are any outliers.

Question 19. Collin sampled 80 students at his school about the number of pets their household had. This data is shown in the following table:

Number of pets	Frequency
0	7
1	31
2	21
3	8
4	6
5	5
6	2

Calculate the mean number of pets per household, and the standard deviation.

Question 20. Alexis surveyed 95 colleagues at her company about the number of projects they were working on. This data is shown in the following table:

Number of projects	Frequency
0	3
1	34
2	48
3	10

Calculate the mean number of projects per employee, and the standard deviation.

Question 21. Sofia sampled 60 properties listed on an estate agent's website and recorded the number of bedrooms the properties had. This data is shown in the following table:

Number of bedrooms	Frequency
1	12
2	18
3	21
4	6
5 to 7	2
8 to 10	1

a) Calculate an estimate for the mean number of bedrooms.
b) Find the median number of bedrooms.

Question 22. The following data was recorded:

x	Frequency
$14 \leq x < 18$	100
$18 \leq x < 22$	200
$22 \leq x < 30$	250
$30 \leq x \leq 32$	50

a) Calculate an estimate for the mean of x
b) Calculate an estimate for the standard deviation of x
c) State the assumption that explains why the midpoints of the class intervals are used to estimate the mean and standard deviation.

Question 23. The weights, in kg, of 30 swans kept on a nature reserve was recorded:

Weight	Frequency
5 to 8	5
9 to 11	15
12 to 13	8
14 to 15	2

a) State the upper class boundary of the “9 to 11" class.
b) Calculate an estimate for the mean weight of the swans.
c) Calculate an estimate for the standard deviation of the weights of the swans.

Question 24. The ages, in years, of 90 teachers working at a sixth form college was recorded:

Age	Frequency
22 to 26	12
27 to 34	30
35 to 45	22
46 to 54	19
55 and over	7

a) State the upper class boundary of the “22 to 26" class.
b) Explain why changing the “55 and over" class to “55 to 66" would be reasonable.
c) Using the class interval suggested in b), calculate estimates for the mean and standard deviation of the ages of the teachers.

Question 25. A publishing company based in London employs 17 members of staff. Their salaries, in thousands of pounds, are recorded in the table below.

Salary	Frequency
35 –	3
40 –	7
45 –	4
50 –	1
55 – 75	2

a) Estimate the mean of the company's salaries.
b) Estimate the standard deviation of the company's salaries.
c) Explain why your answers to a) and b) are only estimates.
d) Explain why your answer to a) should not be publicised as the average wage of people who work in publishing across the UK.

Question 26. Given that $\bar{x} = 9, \sigma^2 = 2, n = 33$, find $\sum x$ and $\sum x^2$

Question 27. Given that $\bar{x} = 90.2, \sigma^2 = 10.96, n = 100$, find $\sum x$ and $\sum x^2$

Question 28. Ebba takes a contract with a company that makes A-Level Maths resources. Her contract says that the average number of hours working per week should be approximately 4, although they will pay her for each hour worked up to a maximum of 8 hours a week. Some projects take Ebba longer than this. Ebba records the number of hours, t, she works each week in the table below.

Hours Worked (t)	Frequency
$0 \leq t < 2$	12
$2 \leq t < 4$	17
$4 \leq t < 6$	21
$6 \leq t < 8$	7
$8 \leq t < 12$	2

Ebba is paid £15.60 an hour for her work. Estimate the mean gross pay she earns from the company each week.

Question 29. Given that $\sum x = 40000, \sum x^2 = 80000, \sigma = 1$, find n

Question 30. The following data has a mean of 30 and a sample standard deviation of 8.70 to 3 significant figures.

19	21	26	27	29	35	39	44

One piece of data is removed and replaced with a new value. The mean increases to 32 but the sample standard deviation decreases. Identify which of the pieces of data could have been removed and find the values they would need to be replaced with.

Question 31. It is given that $\bar{x} = k, \sigma = 5$ and $n = 10$. Two more pieces of data are added, $x = k - 1$ and $k + 1$. Determine the standard deviation of the 12 pieces of data.

Question 32. It is given that $\bar{x} = 200, \sigma = k$ and $n = 10$. Two more pieces of data are added, $x = 2 + k$ and $2 - k$ so that σ is now 99. Determine the value of k

Question 33. On a rainy day, 23 students sit a test. Their teacher marks their papers and finds there to be a mean score of 30 and a standard deviation of 5. Due to the rain, Tom was late to school and missed the test, and so had to complete it later. Tom scored 35. Determine the mean and standard deviation of the scores of all 24 students.

Question 34. On a sunny day, 25 students sit a test. Their teacher marks their papers and finds there to be a mean score of 40 and a standard deviation of 8. The teacher finds that one of his students has cheated and copied another's work. The student's score was 52 and the teacher wants to remove this score from the analysis. Determine the mean and standard deviation of the scores of these remaining 24 students.

Question 35. Linear Coding The data set, x, shown below is coded using $y = \frac{x+5}{4}$

x	15	85	155	295	455

Write down the data set y

Question 36. The values x_i have a mean of $\bar{x} = 85$ and standard deviation $\sigma_x = 12$. The data is coded to the values y_i such that $y = 10x$
Find $\bar{y}$ and σ_y, the mean and standard deviation respectively of the values y_i

Question 37. The values x_i have a mean of $\bar{x} = 120$ and standard deviation $\sigma_x = 5$. The data is coded to the values y_i such that $y = 3x + 50$
Find $\bar{y}$ and σ_y, the mean and standard deviation respectively of the values y_i

Question 38. The values x_i have a mean of $\bar{x}$ and standard deviation σ_x. The data is coded to the values y_i such that $y = 8x - 10$. It is given that $\bar{y} = 500$ and $\sigma_y = 120$
Find $\bar{x}$ and standard deviation σ_x

Question 39. The values x_i have a mean of $\bar{x}$ and standard deviation σ_x. The data is coded to the values y_i such that $y = \frac{20-3x}{9}$. It is given that $\bar{y} = 250$ and $\sigma_y = 45$. Find $\bar{x}$ and standard deviation σ_x

Question 40. The temperatures (°C) at midday for a week in May in Valletta, Malta are recorded in the table below.

24	23	25	27	29	26	24

a) Find the mean and standard deviation of the temperatures in °C
b) The formula to convert from °C to °F is given by $F = \frac{9C}{5} + 32$
Use this formula to determine the mean and standard deviation of the temperatures in °F

Question 41. The temperatures (°F) at midday for a week in May in Southampton, UK are recorded in the table below.

61	63	59	59	64	63	61

a) Find the mean and standard deviation of the temperatures in °F

b) The formula to convert from °F to °C is given by $C = \frac{5(F-32)}{9}$
Use this formula to determine the mean and standard deviation of the temperatures in °C

Question 42. The Mid-Range
For the following data, find the mid-range.

28	35	42	59	60	75

Question 43. Colm teaches Biology to four classes. He set some homework for his students on an online platform and his dashboard recorded how often his students logged in during the week to work on it. This data is shown in the following table:

Log-ins	Frequency
0	12
1	42
2	27
3	10
4	8
5	3
6	1

a) State the mode of the data.
b) State the mid-range of the data.
c) State the median of the data.

Question 44. Linear Interpolation
The following data was recorded:

x	Frequency
$0 \leq x < 5$	35
$5 \leq x < 10$	60
$10 \leq x < 15$	15
$15 \leq x \leq 20$	10

Use linear interpolation to calculate an estimate for the median of x

Question 45. The following data was recorded:

x	Frequency
$0 \leq x < 8$	82
$8 \leq x < 12$	74
$12 \leq x < 20$	62
$20 \leq x \leq 24$	44

Use linear interpolation to calculate an estimate for the median of x

Question 46. The following data was recorded:

x	Frequency
$10 \leq x < 30$	55
$30 \leq x < 70$	95
$70 \leq x < 80$	150
$80 \leq x < 110$	75
$110 \leq x \leq 140$	25

Use linear interpolation to calculate an estimate for the median of x

Question 47. The following data was recorded:

x	Frequency
$50 \leq x < 60$	12
$60 \leq x < 70$	16
$70 \leq x < 80$	28
$80 \leq x < 100$	32
$100 \leq x \leq 120$	14

Use linear interpolation to calculate an estimate for the interquartile range of x

Question 48. The following data was recorded:

x	Frequency
$0 \leq x < 15$	48
$15 \leq x < 40$	93
$40 \leq x < 55$	117
$55 \leq x < 70$	152
$70 \leq x \leq 95$	146
$95 \leq x < 110$	121
$110 \leq x < 135$	94
$135 \leq x < 150$	29

a) Use linear interpolation to calculate an estimate for 20th to the 80th interpercentile range of x
b) Estimate the number of data values that fall within the 20th to the 80th interpercentile range.

Question 49. Andrew measures the acidity of the soil in several randomly selected locations at a nursery. The following data was recorded:

x (pH)	Frequency
$6 \leq x < 6.2$	12
$6.2 \leq x < 6.4$	27
$6.4 \leq x < 6.6$	38
$6.6 \leq x < 6.8$	22
$6.8 \leq x < 7.0$	9

Use linear interpolation to calculate an estimate for the upper quartile of the pH levels recorded.

Question 50. Freya measures the heights, in metres, of a random sample of oak trees in a larger forest. The following data was recorded:

Height (m)	Frequency
13 to 15	7
16 to 18	15
19 to 21	23
22 to 24	9
25 to 26	2

Use linear interpolation to calculate an estimate for the lower quartile of the heights of the oak trees recorded.

Question 51. Lisa runs a carp farm that has over 50 outdoor ponds dedicated to stocking fisheries in the UK. In one pond, there are a large number of young carp. Lisa collects a sample and measure their lengths in centimetres. The following data was recorded:

Length (cm)	Frequency
6 to 7	56
8 to 9	84
10 to 11	129
12 to 13	76
14 to 15	43

Use linear interpolation to calculate an estimate for the length 70% of the fish are longer than.

Question 52. Gerald records the ages, in years, of the members of a local choir. The following data was recorded:

Age (years)	Frequency
25 to 30	9
31 to 40	32
41 to 50	19
51 to 65	26
66 to 80	12

Use linear interpolation to calculate an estimate for the median age of the choir.